Electrical AC BOM (Phase 1)

As-of date: `2026-09-02`

Purpose: maintain the installed Phase 1 AC architecture baseline: portable `30A` EMS, `30A` shore inlet/locking cord with modular `25 ft` TT-30 extension, one `6-way` DIN enclosure, `30A` AC-in breaker, `30A` AC-out main breaker, and two active `20A` GFCI-protected AC-out branches. Owner report `2026-09-02`: all `120V` work is in enclosures, including four duplex devices/eight plug positions total; the system has seen regular use; and the MultiPlus is grounded to truck chassis. Physical enclosure count and chassis-ground installation are closed. Final safety acceptance still requires explicit `LINE/LOAD/PE` proof, downstream protection, GFCI trip/reset in inverter and accepted-shore modes, polarity/neutral-isolation checks, inspection/continuity proof of the installed MultiPlus bond, the separate shell bond, and representative-load verification.

Related docs:

- `docs/implementation/ELECTRICAL_overview_diagram.md`
- `docs/core/SYSTEMS.md`
- `docs/core/TRACKING.md`
- `bom/bom_estimated_items.csv`

Historical provenance

- [INSTALL_MINUS_12_READINESS_PLAN] preserves the May 7 install-window AC planning context; this file owns current Phase 1 AC procurement/implementation.

Historical physical shore-inlet install gate (`2026-07-19`) — closed/superseded by as-built state

- The L5-30 inlet was then purchased but not yet cut into the camper/bed-side interface.
- The electrical module was hard-mounted. The MultiPlus and combined AC breaker enclosure had been removed for the lift and still required remounting at that time.
- From the remounted AC-in breaker endpoint, mock the full `10/3` path to the candidate exterior inlet with required bend radius, cable support, drip/water management, service loop, strain relief, and access to both sides of the cut.
- Any small cable opening through the plywood electrical backer requires a correctly sized grommet/bushing or gland plus independent cable support/strain relief; bare `10/3` must not bear on a raw plywood edge.
- Confirm the exterior cover swing, connector clearance, wall/backing thickness, hidden structure/no-drill zone, butyl bedding land, finish-seal geometry, and a path that does not share the wet-side chase.
- Cut only after the complete inside and outside route is proven. Do not let the easiest exterior location create an inaccessible cable entry or service-obscuring bend behind the electrical module.

Locked AC Architecture

AC-in chain (shore to inverter)

- `shore source/adapters -> portable 30A EMS -> optional 25 ft TT-30 extension -> 30A locking shore cord -> L5-30 shore inlet -> combined 6-way AC DIN enclosure -> 30A UL489 AC-in breaker/disconnect -> MultiPlus AC-in (L/N/PE)`
- AC-in conductors are `10 AWG` / `10/3` on the protected AC-in path (`30A` hardware basis).
- MultiPlus input current limit is set to actual source when adapters are used. Use `10A` for first household tests and `12A` maximum policy on a normal `15A` outlet; do not leave the limit at `50A` on adapter/household shore.
- Use the modular extension only when the extra reach is needed. Fully seat, elevate, and weather-protect the TT-30 midpoint connection, and uncoil both cords under load.

AC-out chain (inverter-backed branch distribution)

- `MultiPlus AC-out-1 -> 10/3 feeder -> combined 6-way AC DIN enclosure -> 30A UL489 AC-out main breaker -> 20A Branch A + 20A Branch B -> GFCI receptacle per branch`
- The two first-in-chain receptacles are `20A` self-test GFCIs: Branch A = office/driver side, Branch B = Galley/passenger side. Owner report `2026-09-02`: each GFCI feeds one downstream duplex from its `LOAD` side, all four devices are enclosed, and AC has seen regular use. Formal electrical acceptance remains open only for the explicit safety/functional checks below.
- Do not reopen the box-count task. Confirm `12 AWG`, `LINE/LOAD/PE`, PE continuity, labels, GFCI trip/reset, whole-chain downstream protection, polarity/neutral isolation, and representative loads in inverter and accepted-shore modes.
- AC-out branch hardware is not required to perform the initial AC-in-only battery charging test, but it is now included in the purchased Phase 1 cart.

Neutral and ground handling

- AC-in and AC-out neutral termination paths remain isolated.
- Separate AC-in neutral pass-through/termination and AC-out neutral bar are required inside the combined enclosure.
- Common equipment grounding/PE bus is acceptable; do not use it as a neutral.
- Continuous equipment ground path and chassis bond are required end-to-end. The installed MultiPlus-II `48/3000/35-50 120V` is a Class I device; its official manual identifies the external `M6` connection as the primary PE point, requires at least `4 mm²` grounding conductor, and explicitly requires the casing to be connected to the vehicle chassis in a mobile shore-power installation. Owner report `2026-09-02`: the MultiPlus is grounded to truck chassis. Verify at the next de-energized inspection that the installed bond originates at the external `M6 PE` lug, uses at least `4 mm²` / selected `10 AWG` green stranded copper with suitable terminations, and lands at a sound chassis point; do not use the aluminum Hiatus shell or 80/20 as the only protective-earth path.
- Bond the aluminum camper shell separately into the same chassis/equipment-ground network with a corrosion-compatible connection. Treat shell hardware/body contact as unproven until low-resistance continuity is measured; use compatible/tinned terminals and antioxidant/isolation practice appropriate to the aluminum interface.
- Do not jumper the MultiPlus case/PE lug to Lynx negative or the `12V` negative bus. The `48V` and `12V` DC return paths remain on their dedicated conductors/buses; Mechman case-ground behavior is a separate commissioning check that may establish the single deliberate house-negative-to-chassis reference through the alternator and its dedicated `2/0` return.
- Do not add an always-bonded downstream neutral-ground bond in branch receptacle wiring. Leave the MultiPlus internal ground relay enabled for the normal single-unit topology: it bonds AC-out neutral to chassis in inverter mode and opens that bond when shore AC is accepted.

AC-out-2 policy

- `AC-out-2` is **reserve-only** in Phase 1.
- Keep labeled panel space and capped route only; no energized branch hardware is procured for this path in Phase 1.

Required Purchasable Components (Phase 1)

Immediate AC-in-only initial charge path

These rows unblock safe MultiPlus shore charging and should not wait on final receptacle count:

Component class	Qty	Rating/listing requirement	BOM row(s)	Phase 1 status
Shore inlet	`1`	`30A` `125V` L5-30 shore inlet with exterior cover	`107`	Purchased
Shore cord, modular extension + household adapter	`1` locking cord + `1` extension + `1` dogbone	`30A` TT-30P-to-L5-30R cord, `25 ft` TT-30P-to-TT-30R extension, and `15A` household dogbone adapter	`108`, `340`, `281`	Purchased
Portable EMS/surge protector	`1`	Portable `30A` EMS with open-neutral/polarity/voltage protection	`123`	Purchased
AC-in breaker/disconnect	`1 of 2`	DIN-mount `UL 489` `1-pole 30A 120VAC`	`13`	Purchased
Combined AC DIN enclosure and bus hardware	`1` enclosure + accessories	`6-way` DIN enclosure; physically isolate AC-in and AC-out neutrals	`109`, `14`	Purchased
Shore + AC-in feed cable	as routed	`10/3` stranded tinned-copper cable for `C-28/C-29` (`30A` path)	`114`	Purchased
Shore-inlet seal/bedding consumables	per install	Butyl bedding + compatible exterior polyurethane perimeter/finish seal	`179`, `180`	Purchased
Strain relief/grommets/clamps/labels/ferrules	per entries	Sized for selected cable/enclosure terminals	`38`, `41`, `43`, `44`, `45`	Cable glands on hand; ferrule kit purchased

Full AC-out distribution / final Phase 1 branch hardware

Component class	Qty	Rating/listing requirement	BOM row(s)	Phase 1 status

Shore inlet	1	30A 125V L5-30 shore inlet with exterior cover	107	Purchased
Shore cord, modular extension + household adapter	1 locking cord + 1 extension + 1 dogbone	30A TT-30P-to-L5-30R cord, 25 ft TT-30P-to-TT-30R extension, and 15A household dogbone adapter	108, 340, 281	Purchased
Portable EMS/surge protector	1	Portable 30A EMS used source-side before the shore cord	123	Purchased
AC-in breaker/disconnect	1	DIN-mount UL 489 1-pole 30A 120VAC	13	Purchased
AC-out main breaker	1	DIN-mount UL 489 1-pole 30A 120VAC, ahead of branch breakers	327	Purchased
Combined DIN enclosure	1	6-way DIN enclosure with slot plan: 30A in, 30A out, 20A, 20A, 2x blank/spare	109	Purchased
AC-out branch breaker set	2 active	DIN-mount UL 489 branch breakers: 20A + 20A	110, 330	Purchased
DIN accessory kit	1 kit	Neutral isolation hardware, PE/ground bus support, ferrules, labels/blanks as needed	14, 41	Purchased / labels-blanks confirm on hand
MultiPlus/chassis and shell bonding materials	field-measured	Owner confirms MultiPlus-to-truck-chassis ground installed; verify M6 PE, conductor, termination, endpoint, and continuity. Separate shell bond jumper; M6/tinned lugs, protected chassis attachment, compatible aluminum-interface hardware/compound	51, 339	Chassis ground installed; shell bond inventory/buy
GFCI receptacles	2	20A self-test GFCI receptacles, one per active branch	15	Purchased
Standard downstream receptacles	2 installed / owner-reported	One downstream duplex from each first GFCI LOAD, same 12 AWG branch, accessible listed box	inactive historical 111; installed inventory source not yet reconciled	Inspect/test as-built
Outlet boxes + covers/faceplates + clamps	4 installed device sets / owner-reported	Six 14 cu in old-work boxes were purchased; prove each used device's listed box, compatible cover/plate and clamp, conductor fill, wall-stack fit, and service access	active row 112 plus inventory	Physical acceptance open
AC branch cable	30 ft purchased	12/3 stranded triplex branch cable (C-31/C-32)	113	Purchased
Shore + AC-in/AC-out feeder cable	20 ft purchased	10/3 stranded triplex for C-28/C-29/C-30 (30A paths)	114	Purchased
Strain relief/cable glands	per enclosure entries	Use assorted on-hand entry hardware sized during physical layout	44	On-hand fitment stock
Grommets	per pass-through points	Abrasion protection at penetrations, selected hands-on during routing	43	On-hand / install fit
P-clamps and retention hardware	per route	Cable support and vibration control, selected hands-on during routing	45	On-hand / install fit
Loom/sleeving	per exposed runs	Harness abrasion protection	42	Required
Heat shrink (adhesive)	install consumable	Termination sealing and strain relief support	38	Required

Ferrules/terminals (AC-relevant)	install consumable	Sized to `10 AWG` and `12 AWG` terminations as required by device terminals	`41`, `116`	Required
AC-out-2 branch breaker/protection hardware	`0` in Phase 1	Reserve-only route, no energized branch hardware in this phase	N/A	Reserve-only (not procured)

First-Live AC-In Test Result (`2026-05-27`)

- AC Input 1 should be labeled `Shore power` for this mobile/source-current-limited system.
- MultiPlus switch `II` is charger-only; switch `I` is inverter/charger normal mode; `0` is off.
- Short shore test passed with household-source current limiting: about `1294W` shore input and about `54.3V x 21.6A` (`~1173W`) battery charge in bulk.
- Current disconnect practice: connect RV/EMS/load side first, then energize shore; disconnect in reverse by de-energizing/unplugging shore source before disconnecting the RV side.
- This result does not close battery-charge-profile commissioning. Program/verify the MultiPlus lithium settings before sustained charging.

Manual AC Validation Checklist

0) AC-in-only initial charger validation

- Confirm AC-in physical order: `shore source/adapters -> portable EMS -> optional TT-30 extension -> locking shore cord -> inlet -> AC-in breaker/disconnect -> MultiPlus AC-in`.
- Confirm AC-out breakers/loads are disconnected or not yet installed for the first battery-charge test.
- Confirm MultiPlus input current limit is set to actual source (`10A` first household test, `12A` max on normal `15A`, actual rating for `20A` / `30A`).
- Confirm AC Input 1 is labeled/configured as `Shore power`.
- Confirm battery charge profile is intentionally set for the selected LiFePO4 voltage basis before sustained charging.
- Confirm AC-in and AC-out neutral paths are not mixed.

Use this checklist as the continuing acceptance gate before sustained shore charging, AC-out branch energization, or any AC enclosure closeout.

1) Topology integrity

- Confirm one unique AC-in chain exists: `shore source/adapters -> portable EMS -> cord -> inlet -> AC-in breaker -> MultiPlus AC-in`.
- Confirm one unique AC-out-1 chain exists: `MultiPlus AC-out-1 -> 30A AC-out main -> 20A branch breakers -> GFCI receptacles`.
- Confirm `AC-out-2` is documented as reserve-only and not active in Phase 1 procurement.
- Confirm the two first-in-chain GFCIs remain the only branch-origin devices and each owner-reported downstream duplex is fed only from its corresponding GFCI `LOAD`; prove compatible listed boxes/covers/clamps, conductor fill/accessibility, and whole-chain testing.

2) Protection coordination

- Confirm AC-in breaker is `30A` and AC-in conductors are `10 AWG`.
- Confirm AC-out main breaker is `30A` and the MultiPlus-to-enclosure feeder is `10 AWG`.
- Confirm branch OCP values are `20A` and `20A` with `12 AWG` branch conductors.
- Confirm breaker listing basis is `UL 489` (or equivalent NRTL listing) for branch/feeder use.

3) Neutral/ground correctness

- Confirm AC-in and AC-out neutral paths are isolated.
- Confirm input and output neutral paths are separately landed/passed through inside the combined enclosure.
- Confirm the equipment grounding/PE path is common and continuous.
- Confirm the MultiPlus external `M6 PE` lug is bonded to a verified truck-chassis point with at least `4 mm²` / selected `10 AWG` green stranded copper; the aluminum shell/80/20 is not the sole return.
- Confirm the aluminum shell has its own corrosion-compatible bond to the chassis/equipment-ground network and verify low-resistance continuity after final assembly.
- Confirm no intentional jumper exists from MultiPlus case/PE to Lynx negative or the `12V` negative bus, and confirm no chassis path bypasses the SmartShunt after the Mechman ground style is physically identified.
- Confirm no fixed downstream neutral-ground bond is added in branch receptacle wiring.
- Confirm the MultiPlus internal ground relay remains enabled for this normal single-unit mobile topology; test both GFCI branches in inverter mode and on accepted shore power before routine use.

4) Procurement completeness

- Confirm every required AC component class has a BOM row mapping.
- Confirm no AC-critical component exists only as implied text.
- Confirm AC-out-2 hardware remains excluded from Phase 1 carts.

5) Documentation parity

- Confirm AC assumptions match across:
  - `docs/implementation/ELECTRICAL_AC_BOM.md`
  - `docs/implementation/ELECTRICAL_overview_diagram.md`
  - `docs/core/SYSTEMS.md`
  - `docs/core/TRACKING.md`
  - `bom/bom_estimated_items.csv`

6) Operating scenarios

- AC-in-only initial charge: no AC-out loads connected, MultiPlus charger behavior documented.
- Shore present (`30A` source): pass-through + charging behavior documented.
- Shore present via `15A/20A` adapter: current-limit setting policy documented.
- Shore absent: inverter-backed `AC-out-1` behavior documented.
- GFCI trip/reset behavior per branch is called out for commissioning test.

Procurement Notes

- DIN rail is a mounting method; breaker listing and rating remain the controlling requirement.
- Lowest-cost listed policy is acceptable only if each selected device has verifiable NRTL listing (`UL` or `ETL`) for intended use.
- Purchased SKU lock is recorded in `bom/bom_estimated_items.csv` for rows `13` , `327` , `14` , `15` , `41` , `107` , `108` , `109` , `110` , `112` , `330` , `113` , `114` , `123` , `179` , and `180` . Rows `13` / `327` are the AC-in/AC-out `30A` pair; rows `110` / `330` are the two `20A` branch breakers. Row `112` is the purchased old-work box six-pack, not evidence that six downstream devices are approved. Row `181` was same-order tire-deflator/off-road support hardware and is intentionally outside AC scope.
- Inactive BOM row `111` preserves the formerly closed downstream-receptacle concept. Row `112` supplied the six purchased old-work boxes. Owner report `2026-08-27` closes physical device count at two GFCIs plus two downstream duplexes; installed receptacle/cover/clamp inventory provenance and all box-fill/wall-stack/accessibility/testing gates remain open.
- Current utilization note (`2026-05-16`): AC protection chain and purchase scope are locked for a `30A` system with two active `20A` branches. Do not add a third active AC branch without revisiting the AC-out main/enclosure/feed plan.